H524 Introduction to Biostatistics

Week 6: Power, Two-Sample Tests, and ANOVA

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Fall 2025

Outline

1. Review from Week 5
2. Statistical Power (Chapter 11)
3. Two-Sample t-Tests: Independent Samples (Chapter 12)
4. Paired t-Tests (Chapter 12)
5. Analysis of Variance (ANOVA) (Chapter 12)
6. AI-Enhanced Learning for Power and Comparisons
7. Summary and Key Formulas

Quick Review: Hypothesis Testing

Last week we covered:

- Null and alternative hypotheses
- Test statistics and p-values
- One-sample t-tests for means
- Type I error (false positive) and significance level α
- Rejection regions and critical values

This Week: Extending hypothesis testing

- Statistical power and Type II errors
- Comparing two independent samples
- Paired samples and dependent data
- Analysis of Variance (ANOVA) for multiple groups

Reading: Pagano & Gauvreau Chapters 11-12

Hypothesis Testing Errors

Two types of errors in hypothesis testing:

Decision	**Truth** H_0 True	 H_0 False
Reject H_0	Type I Error (False Positive) Probability $= \alpha$	Correct Decision (True Positive) Probability $= 1 - \beta$ (Power)
Fail to Reject H_0	Correct Decision (True Negative) Probability $= 1 - \alpha$	Type II Error (False Negative) Probability $= \beta$

Key Definitions:

- α = Type I error rate (significance level, usually 0.05)
- β = Type II error rate (probability of missing real effect)
- **Power** $= 1 - \beta$ (probability of detecting real effect)

What is Statistical Power?

Definition: Power is the probability of correctly rejecting H_0 when it is false.

$$\text{Power} = 1 - \beta = \Pr(\text{Reject } H_0 \mid H_0 \text{ is false})$$

Interpretation:

- Power $= 0.80$ means 80% chance of detecting a real effect
- If power is too low, you might miss important findings
- Typically want power ≥ 0.80 (80%)
- Some fields require power ≥ 0.90 (90%)

Example:

- Testing a new drug vs. placebo
- Drug truly reduces blood pressure by 10 mmHg
- Power $= 0.80$ means 80% chance our study will detect this effect
- 20% chance we'll conclude "no effect" (Type II error)

Factors Affecting Power

Power increases with:

1. **Larger Sample Size (n)**
   - More data $\Rightarrow$ smaller standard error $\Rightarrow$ easier to detect effects
   - Most important and controllable factor
2. **Larger Effect Size ($|\mu - \mu_0|$ or $|\mu_1 - \mu_2|$)**
   - Bigger differences are easier to detect
   - Not under researcher's control
3. **Lower Variability (σ)**
   - Less noise $\Rightarrow$ clearer signal
   - Can be reduced by careful measurement and study design
4. **Higher Significance Level (α)**
   - Trade-off: higher α increases power but increases Type I error
   - Usually keep $\alpha = 0.05$ fixed

Power Calculation Example

Scenario: Testing if a new drug lowers cholesterol

Given:

- Population mean cholesterol: $\mu_0 = 200$ mg/dL
- We want to detect a reduction to $\mu = 190$ mg/dL
- Known SD: $\sigma = 30$ mg/dL
- Sample size: $n = 25$
- Significance level: $\alpha = 0.05$ (two-sided)

Question: What is the power to detect this 10 mg/dL reduction?

Solution:

1. Standard error: $\text{SE} = 30/\sqrt{25} = 6$ mg/dL
2. Critical value: $z_{0.975} = 1.96$
3. Rejection region: Reject if $\bar{x} < 200 - 1.96(6) = 188.24$ or $\bar{x} > 211.76$
4. Under true $\mu = 190$: Calculate $\Pr(\bar{X} < 188.24 \text{ or } \bar{X} > 211.76 \mid \mu = 190)$

Power Calculation in R

Calculate power for the cholesterol example:

```
# Parameters
mu0 <- 200      # Null hypothesis mean
mu1 <- 190      # True mean (alternative)
sigma <- 30     # Population SD
n <- 25         # Sample size
alpha <- 0.05   # Significance level

# Standard error
se <- sigma / sqrt(n)  # 6

# Critical values (two-sided)
z_crit <- qnorm(1 - alpha/2)  # 1.96
lower_crit <- mu0 - z_crit * se  # 188.24
upper_crit <- mu0 + z_crit * se  # 211.76

# Power: P(reject H0 | mu = 190)
power_lower <- pnorm(lower_crit, mean=mu1, sd=se)
```

Power Calculation Results

Result: Power $= 0.345$ (34.5%)

Interpretation:

- Only 34.5% chance of detecting a 10 mg/dL reduction with $n = 25$
- 65.5% chance of Type II error (missing the effect)
- This is too low! We want power ≥ 0.80

Solution: Increase sample size

For power $= 0.80$, we need approximately $n = 143$ subjects

General lesson:

- Always calculate power **before** conducting study
- Helps determine necessary sample size
- Avoids wasting resources on underpowered studies
- Part of good study design and grant proposals

Sample Size Calculation in R

Using `power.t.test()` **function:**

```
# Calculate required sample size for power = 0.80
power.t.test(delta = 10,      # Effect size (|mu1 - mu0|)
             sd = 30,           # Standard deviation
             sig.level = 0.05, # Significance level
             power = 0.80,     # Desired power
             type = "one.sample")

# Output:
#      One-sample t test power calculation
#
#              n = 142.246
#          delta = 10
#             sd = 30
#      sig.level = 0.05
#          power = 0.8
#
# Need n = 143 subjects
```

Visualizing how power changes with sample size:

[Power increases non-linearly with sample size]

Key observations:

- Power increases rapidly initially
- Diminishing returns for very large n
- To double power from 0.40 to 0.80, need $\sim$4x sample size
- To cut standard error in half, need 4x sample size

Practical implications:

- Small studies ($n < 30$) often severely underpowered
- "Pilot studies" rarely have power to detect effects
- Need careful planning and power analysis before data collection
- Consider effect size: smaller effects need larger samples

Comparing Two Independent Groups

Common research questions:

- Does treatment A work better than treatment B?
- Do men and women differ in average blood pressure?
- Is mean cholesterol higher in smokers vs. non-smokers?

Two-sample problem:

- Two independent groups (treatment/control, male/female, etc.)
- Want to compare population means: μ_1 vs. μ_2
- Observe sample means: $\bar{x}_1$ and $\bar{x}_2$

Hypotheses:

$$H_0 : \mu_1 = \mu_2 \quad \text{(or equivalently: } \mu_1 - \mu_2 = 0\text{)}$$
$$H_A : \mu_1 \neq \mu_2 \quad \text{(two-sided)}$$

Or one-sided: $H_A : \mu_1 > \mu_2$ or $H_A : \mu_1 < \mu_2$

Two-Sample t-Test: Assumptions

Assumptions for validity:

1. **Independence**
   - Observations within each group are independent
   - The two groups are independent of each other
   - Violated by paired data (covered later)
2. **Normality**
   - Data in each group approximately normal
   - Or large sample sizes ($n_1, n_2 \geq 30$) for CLT
   - Check with histograms, Q-Q plots
3. **Equal Variances (for pooled t-test)**
   - $\sigma_1^2 = \sigma_2^2$ (homogeneity of variance)
   - Can test with F-test or Levene's test
   - If violated, use Welch's t-test (does not assume equal variances)

Pooled Two-Sample t-Test

When variances are equal: $\sigma_1^2 = \sigma_2^2 = \sigma^2$

Test statistic:

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - 0}{s_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

where the **pooled standard deviation** is:

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$$

Distribution: Under H_0, $t \sim t_{n_1+n_2-2}$

Degrees of freedom: $df = n_1 + n_2 - 2$

Decision rule:

- Calculate p-value: $p = 2 \times \Pr(T > |t| \mid H_0)$ for two-sided
- Reject H_0 if $p < \alpha$

Example: Tumor Growth Study

Study: Effect of new drug on tumor growth (mm/month)

Control	Treatment
7	4
10	6
9	10
8	8
7	5
6	3
8	10
9	8
12	8
13	10

Question: Does the treatment reduce tumor growth?

Hypotheses:

Two-Sample t-Test in R

```
# Enter data
control <- c(7, 10, 9, 8, 7, 6, 8, 9, 12, 13)
treatment <- c(4, 6, 10, 8, 5, 3, 10, 8, 8, 10)

# Summary statistics
mean(control)    # 8.9 mm
mean(treatment)  # 7.2 mm
sd(control)      # 2.234 mm
sd(treatment)    # 2.573 mm

# Pooled two-sample t-test (equal variances)
t.test(control, treatment,
       var.equal = TRUE,     # Assume equal variances
       alternative = "greater")  # One-sided: control > treatment

# Output:
# t = 1.571, df = 18, p-value = 0.0671
# mean of x = 8.9, mean of y = 7.2
# 95% CI: (-0.182, Inf)
```

Welch's t-Test (Unequal Variances)

When variances are unequal: $\sigma_1^2 \neq \sigma_2^2$

Test statistic:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Degrees of freedom (Welch-Satterthwaite):

$$df = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}}$$

In R: Just use `t.test()` with `var.equal = FALSE` (default!)

When to use:

- Always safer to use Welch's test
- More robust, similar power when variances equal
- Default in R's `t.test()` function

Confidence Interval for Difference in Means

95% CI for $\mu_1 - \mu_2$ (pooled):

$$(\bar{x}_1 - \bar{x}_2) \pm t_{0.975,df} \times s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

where $df = n_1 + n_2 - 2$

Tumor example:

- $\bar{x}_C - \bar{x}_T = 8.9 - 7.2 = 1.7$ mm
- $s_p = 2.405$ mm
- $t_{0.975,18} = 2.101$
- $\text{SE} = 2.405\sqrt{\frac{1}{10} + \frac{1}{10}} = 1.076$ mm
- 95% CI: $1.7 \pm 2.101 \times 1.076 = 1.7 \pm 2.26$
- **95% CI: (-0.56, 3.96) mm**

Interpretation: We are 95% confident the true difference is between -0.56 and 3.96 mm. Since interval includes 0, no significant difference at $\alpha = 0.05$.

Paired vs. Independent Samples

Paired (dependent) samples:

- Same subjects measured twice (before/after)
- Matched pairs (twins, siblings, matched cases)
- Left vs. right measurements on same subject

Examples:

- Blood pressure before and after medication
- Weight before and after diet intervention
- Pain score before and after treatment
- Test scores with two different methods on same students

Why pairing matters:

- Removes between-subject variability
- More powerful than two-sample t-test
- Controls for confounding variables
- **Mistake:** Using two-sample test on paired data loses power

Paired t-Test Procedure

Key idea: Analyze the *differences*

For each pair i:

$$d_i = x_{i,\text{before}} - x_{i,\text{after}}$$

or $d_i = x_{i,1} - x_{i,2}$

Then: Perform one-sample t-test on the differences

Hypotheses:

$$H_0 : \mu_d = 0 \quad \text{(no change)}$$
$$H_A : \mu_d \neq 0 \quad \text{(two-sided)}$$

Test statistic:

$$t = \frac{\bar{d} - 0}{s_d / \sqrt{n}}$$

where $\bar{d}$ = mean difference, s_d = SD of differences, n = number of pairs

Distribution: Under H_0, $t \sim t$

Example: Blood Pressure Medication

Study: 10 patients, blood pressure measured before and after medication

Patient	Before	After	Difference (d_i)
1	142	138	4
2	138	132	6
3	150	148	2
4	148	140	8
5	135	135	0
6	160	152	8
7	155	148	7
8	145	142	3
9	152	146	6
10	158	150	8
Mean	148.3	143.1	5.2
SD	8.23	6.86	2.78

Question: Does medication significantly reduce blood pressure?

Paired t-Test in R

```r
# Enter data
before <- c(142, 138, 150, 148, 135, 160, 155, 145, 152, 158)
after <- c(138, 132, 148, 140, 135, 152, 148, 142, 146, 150)

# Calculate differences
differences <- before - after
mean(differences)  # 5.2 mmHg
sd(differences)    # 2.78 mmHg

# Paired t-test (Method 1: on differences)
t.test(differences, mu = 0, alternative = "greater")

# Paired t-test (Method 2: paired argument)
t.test(before, after, paired = TRUE, alternative = "greater")

# Output:
# t = 5.916, df = 9, p-value = 0.0001236
# mean of differences = 5.2
# 95% CI: (3.43, Inf)
```

Paired vs. Independent t-Test Comparison

What if we (incorrectly) used two-sample test?

Using two-sample t-test: $p = 0.115$ (not significant!)
Using paired t-test: $p = 0.0001$ (highly significant!)

Why such a big difference?

- Pairing removes between-subject variability
- Standard error is much smaller: $\text{SE} = 2.78/\sqrt{10} = 0.88$
- Two-sample SE: $\sqrt{8.23^2/10 + 6.86^2/10} = 3.25$ (nearly 4x larger!)
- Pairing gives much more power

Lesson:

- Always use paired test for paired data
- Pairing is a powerful design strategy
- Can dramatically reduce sample size needed

Comparing More Than Two Groups

Limitation of t-tests: Only compare two groups at a time

What if we have 3+ groups?

- Compare 3 drug doses: low, medium, high
- Compare 4 treatment methods
- Compare outcomes across multiple hospitals

Naive approach: Multiple t-tests

- Compare groups 1 vs. 2, 1 vs. 3, 2 vs. 3, etc.
- For k groups: $\binom{k}{2} = \frac{k(k-1)}{2}$ comparisons
- For $k = 3$: 3 tests; for $k = 4$: 6 tests; for $k = 5$: 10 tests

Problem: Multiple testing inflation

- Each test has Type I error rate $\alpha = 0.05$
- Overall Type I error rate much higher!
- For 3 independent tests: $1 - (1 - 0.05)^3 = 0.143$ (14.3%)
- Need a method that controls overall error rate

Analysis of Variance (ANOVA)

Solution: One omnibus test for all groups

Hypotheses:

$$H_0 : \mu_1 = \mu_2 = \cdots = \mu_k \quad \text{(all means equal)}$$
$$H_A : \text{At least one mean differs}$$

Key idea: Compare variation *between* groups to variation *within* groups

- **Between-group variation:** How different are the group means?
- **Within-group variation:** How variable are observations within each group?

If H_0 is true:

- All groups sampled from same population
- Between-group variation $\approx$ within-group variation
- Differences in sample means just due to chance

If H_A is true:

ANOVA: Partitioning Variability

Total variability = Between-group + Within-group

Sum of Squares decomposition:

$$SS_{\text{Total}} = SS_{\text{Between}} + SS_{\text{Within}}$$

Total Sum of Squares:

$$SS_{\text{Total}} = \sum_{i=1}^{k} \sum_{j=1}^{n_i} (x_{ij} - \bar{x})^2$$

Between-Group Sum of Squares:

$$SS_{\text{Between}} = \sum_{i=1}^{k} n_i (\bar{x}_i - \bar{x})^2$$

Within-Group Sum of Squares:

$$SS_{\text{Within}} = \sum^{k} \sum^{n_i} (x_{ij} - \bar{x}_i)^2$$

ANOVA: Mean Squares and F-Statistic

Mean Squares = Sum of Squares / degrees of freedom

Between-Group Mean Square:

$$\text{MS}_{\text{Between}} = \frac{\text{SS}_{\text{Between}}}{k-1}$$

Within-Group Mean Square (pooled variance):

$$\text{MS}_{\text{Within}} = \frac{\text{SS}_{\text{Within}}}{n-k}$$

where k = number of groups, n = total sample size

F-Statistic:

$$F = \frac{\text{MS}_{\text{Between}}}{\text{MS}_{\text{Within}}}$$

Distribution: Under H_0, $F \sim F_{k-1,n-k}$

Decision rule:

- Large $F \rightarrow$ evidence against H_0

ANOVA Table

Standard ANOVA output format:

Source	SS	df	MS	F
Between Groups	$SS_{Between}$	$k-1$	$MS_{Between}$	F
Within Groups	SS_{Within}	$n-k$	MS_{Within}	
Total	SS_{Total}	$n-1$		

Interpretation:

- Large F-statistic $\Rightarrow$ between-group variation is large relative to within-group
- Small p-value $\Rightarrow$ reject H_0, conclude at least one mean differs
- If reject H_0: Follow up with post-hoc tests to see which groups differ

Note: ANOVA tells us there's a difference somewhere, but not where!

ANOVA Assumptions

For valid ANOVA results:

1. **Independence**
   - Observations within and between groups are independent
   - Random sampling or random assignment

2. **Normality**
   - Data within each group approximately normal
   - Robust to violations with large samples (CLT)
   - Check with histograms, Q-Q plots for each group

3. **Equal Variances (Homogeneity of Variance)**
   - $\sigma_1^2 = \sigma_2^2 = \cdots = \sigma_k^2$
   - Check with Levene's test or Bartlett's test
   - Or visually: boxplots should have similar spreads
   - If violated: Use Welch's ANOVA or transformation

4. **Balanced Design (preferred but not required)**

Example: Drug Dose Study

Study: Effect of 3 drug doses on cholesterol reduction (mg/dL)

Low Dose	Medium Dose	High Dose
10	15	25
12	18	28
8	16	30
11	20	26
9	17	27
Mean: 10.0 SD: 1.58	Mean: 17.2 SD: 1.92	Mean: 27.2 SD: 1.92

Question: Is there a significant difference in cholesterol reduction across doses?

Hypotheses:

- $H_0 : \mu_{\text{low}} = \mu_{\text{med}} = \mu_{\text{high}}$

ANOVA in R

```
# Enter data
low <- c(10, 12, 8, 11, 9)
medium <- c(15, 18, 16, 20, 17)
high <- c(25, 28, 30, 26, 27)

# Create data frame in long format
cholesterol <- c(low, medium, high)
dose <- factor(rep(c("Low", "Medium", "High"), each=5),
               levels=c("Low", "Medium", "High"))
data <- data.frame(cholesterol, dose)

# Perform ANOVA
model <- aov(cholesterol ~ dose, data=data)
summary(model)

# Output:
#             Df Sum Sq Mean Sq F value   Pr(>F)
# dose         2 985.73  492.87   149.4 7.77e-09 ***
# Residuals   12  39.60    3.30
```

Post-Hoc Tests

After significant ANOVA: Which groups differ?

Problem: Multiple pairwise comparisons inflates Type I error

Solution: Post-hoc tests that control family-wise error rate

Common post-hoc tests:

1. **Tukey's HSD (Honestly Significant Difference)**
   - Most common and powerful
   - Controls overall Type I error at α
   - All pairwise comparisons
2. **Bonferroni Correction**
   - Simple but conservative
   - Use $\alpha^* = \alpha/m$ for each test (m = number of comparisons)
   - For 3 groups: $\alpha^* = 0.05/3 = 0.0167$
3. **Scheffe's Method**

Post-Hoc Tests in R

```
# Tukey's HSD test
TukeyHSD(model)

# Output:
#   Tukey multiple comparisons of means
#     95% family-wise confidence level
#
# Fit: aov(formula = cholesterol ~ dose, data = data)
#
# $dose
#                    diff     lwr     upr     p adj
# Medium-Low       7.2000   4.413  10.0870  0.000032
# High-Low        17.2000  14.413  19.9870  0.000000
# High-Medium     10.0000   7.213  12.7870  0.000001

# All pairwise differences are significant
# High dose reduces cholesterol significantly more than
# medium dose, which reduces more than low dose
```

ANOVA vs. Multiple t-Tests

Why ANOVA instead of multiple t-tests?

Advantages of ANOVA:

1. **Controls Type I error rate**
   - One test at $\alpha = 0.05$ level
   - Multiple t-tests inflate error rate
2. **More powerful**
   - Pools variance estimate across all groups
   - More degrees of freedom for error
3. **Can test complex hypotheses**
   - Test overall effect of categorical variable
   - Can include multiple factors (two-way ANOVA)
4. **Provides comprehensive summary**
   - Single test answers overall question
   - Follow up only if significant

When to use t-test: Only 2 groups to compare

AI Tools for Power Analysis and Testing

How AI can help this week:

- **Power calculations:** Sample size determination
- **Study design:** Choosing between paired and independent designs
- **Test selection:** When to use t-test vs. ANOVA
- **R code generation:** Performing power analysis and tests
- **Assumption checking:** Verifying normality and equal variances
- **Results interpretation:** Understanding ANOVA tables and post-hoc tests

Example prompts:

- "Calculate required sample size to detect 10-point difference in mean with 80% power"
- "Explain when to use paired vs. independent t-test with medical examples"
- "Generate R code for one-way ANOVA with Tukey post-hoc tests"
- "How do I check ANOVA assumptions in R?"

AI Demo: Power Analysis

Prompt: "Calculate sample size needed to detect 5 mmHg reduction in blood pressure with power 0.80, assuming SD=15"

Claude AI Response:

```
# Power analysis for one-sample t-test
# Detect 5 mmHg reduction from baseline
# Assume SD = 15 mmHg, power = 0.80, alpha = 0.05

power.t.test(delta = 5,        # Effect size
             sd = 15,           # Standard deviation
             sig.level = 0.05,  # Alpha
             power = 0.80,      # Desired power
             type = "one.sample")

# Result: n = 142.2 (need 143 subjects)

# Visualize power curve
n_values <- seq(10, 300, by=10)
power_values <- sapply(n_values, function(n) {
  power.t.test(n=n, delta=5, sd=15, sig.level=0.05,
               type="one.sample")$power
})

plot(n_values, power_values, type="l", lwd=2,
     xlab="Sample Size", ylab="Power",
     main="Power vs. Sample Size",
     col="blue")
abline(h=0.80, col="red", lty=2)
abline(v=143, col="red", lty=2)
grid()
```

AI Demo: Choosing the Right Test

Prompt: "I have blood pressure measured before and after treatment in 20 patients. Which test should I use?"

Claude AI Response:

```
You should use a PAIRED t-test because:

1. Same subjects measured twice (before/after)
2. Data are dependent (matched pairs)
3. Want to compare mean change from baseline

# R code:
before <- c(...)  # Your before data
after <- c(...)   # Your after data

# Method 1: Test on differences
differences <- before - after
t.test(differences, mu=0, alternative="two.sided")

# Method 2: Using paired argument
```

AI Demo: ANOVA and Post-Hoc Tests

Prompt: “Perform ANOVA comparing 4 treatment groups and follow up with post-hoc tests in R”

Claude AI Response:

```
# Assume data structure:
# outcome: numeric response variable
# treatment: factor with 4 levels

# Step 1: Check assumptions
# Visualize distributions
boxplot(outcome ~ treatment, data=mydata,
        main="Outcome by Treatment Group")

# Test homogeneity of variance
bartlett.test(outcome ~ treatment, data=mydata)
# If p > 0.05, variances are equal (good!)

# Step 2: Perform ANOVA
model <- aov(outcome ~ treatment, data=mydata)
summary(model)

# Step 3: If ANOVA significant, do post-hoc tests
# Tukey's HSD (most common)
TukeyHSD(model, conf.level=0.95)

# Visualize Tukey results
plot(TukeyHSD(model))

# Interpretation:
# - ANOVA tests if ANY groups differ (omnibus test)
```

AI Best Practices for Hypothesis Tests

DO:

- Check test assumptions first
- Visualize data before testing
- Consider study design (paired?)
- Calculate power before study
- Interpret effect sizes, not just p-values
- Verify AI's test selection

DON'T:

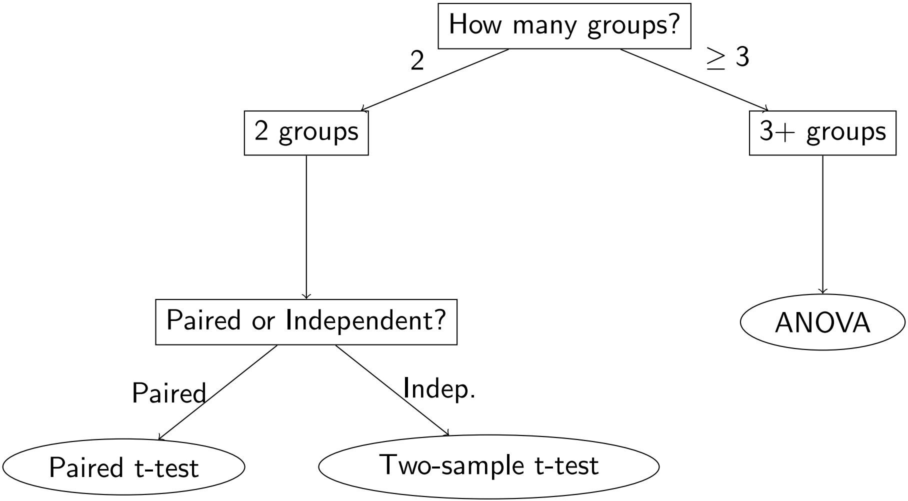
- Use wrong test type (paired vs. independent)
- Run many tests without correction
- Ignore violated assumptions
- Trust AI code blindly
- P-hack by trying many tests
- Forget about power/sample size

Critical thinking checklist:

1. Are observations independent or paired?
2. How many groups am I comparing?
3. Are assumptions met (normality, equal variance)?
4. What is my hypothesis (one-sided vs. two-sided)?
5. Do I need post-hoc tests?

Choosing the Right Test

Decision tree:



Key Formulas: Power and Sample Size

Type I and Type II Errors:

- α = P(Reject H_0 | H_0 true) = Type I error
- β = P(Fail to reject H_0 | H_0 false) = Type II error
- Power = $1 - \beta$ = P(Reject H_0 | H_0 false)

Power depends on:

- Sample size n (larger $\Rightarrow$ more power)
- Effect size $|\mu_1 - \mu_0|$ (larger $\Rightarrow$ more power)
- Variability σ (smaller $\Rightarrow$ more power)
- Significance level α (larger $\Rightarrow$ more power, but more Type I error)

In R:

- `power.t.test()` for one-sample and two-sample
- Specify any 4 of: n, delta, sd, sig.level, power
- Function solves for the missing parameter

Key Formulas: Two-Sample t-Test

Pooled two-sample t-test (equal variances):

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \quad \text{where} \quad s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$$

Distribution: $t \sim t_{n_1+n_2-2}$ under $H_0 : \mu_1 = \mu_2$

Confidence interval for $\mu_1 - \mu_2$:

$$(\bar{x}_1 - \bar{x}_2) \pm t_{\alpha/2, n_1+n_2-2} \times s_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

Welch's t-test (unequal variances):

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

with complicated df formula (R calculates automatically)

Key Formulas: Paired t-Test

Paired t-test:

For differences $d_i = x_{i,1} - x_{i,2}$:

$$t = \frac{\bar{d}}{s_d/\sqrt{n}}$$

where $\bar{d}$ = mean difference, s_d = SD of differences

Distribution: $t \sim t_{n-1}$ under $H_0 : \mu_d = 0$

Confidence interval for μ_d:

$$\bar{d} \pm t_{\alpha/2,n-1} \times \frac{s_d}{\sqrt{n}}$$

When to use:

- Before/after measurements on same subjects
- Matched pairs (twins, siblings, matched controls)
- Left/right measurements
- Any dependent samples

Key Formulas: ANOVA

F-statistic:

$$F = \frac{\text{MS}_{\text{Between}}}{\text{MS}_{\text{Within}}} = \frac{\text{SS}_{\text{Between}}/(k-1)}{\text{SS}_{\text{Within}}/(n-k)}$$

Distribution: $F \sim F_{k-1,n-k}$ under $H_0 : \mu_1 = \cdots = \mu_k$

Sum of squares:

$$\text{SS}_{\text{Total}} = \sum_i \sum_j (x_{ij} - \bar{x})^2$$

$$\text{SS}_{\text{Between}} = \sum_i n_i(\bar{x}_i - \bar{x})^2$$

$$\text{SS}_{\text{Within}} = \sum_i \sum_j (x_{ij} - \bar{x}_i)^2$$

Relationship: $\text{SS}_{\text{Total}} = \text{SS}_{\text{Between}} + \text{SS}_{\text{Within}}$

Post-hoc: If F-test significant, use Tukey's HSD or Bonferroni to identify which groups differ

Summary: This Week

Today we covered:

- **Chapter 11: Power**
  - Type I and Type II errors
  - Statistical power and factors affecting it
  - Sample size calculations
- **Chapter 12: Two-Sample Tests**
  - Independent two-sample t-test (pooled and Welch's)
  - Paired t-test for dependent samples
  - Confidence intervals for differences
- **Chapter 12: ANOVA**
  - Comparing 3+ groups simultaneously
  - F-test and ANOVA table
  - Post-hoc tests (Tukey, Bonferroni)

Next Week: Non-parametric tests (Chapter 13)

Reading: Chapter 13 (Wilcoxon, Kruskal-Wallis, Sign test)

Questions?

Thank you!

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